Production Incident Management Procedure

Document ID	ENG-004
Department	Engineering
Document type	Procedure
Owner	Head of Site Reliability Engineering
Version	2.3
Status	APPROVED
Effective / last reviewed	2025-09-20
Audience	Engineering staff
Related documents	ENG-005, ENG-010, SEC-010, CS-001, CS-005

Summary. Defines how Northstar Systems detects, classifies, runs, communicates and reviews production incidents. Covers SEV1 to SEV4 definitions, response targets, incident roles, status page updates, escalation, post-incident reviews and worked examples.

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1. Purpose and Scope

This procedure defines how Engineering at Northstar Systems Private Limited detects, classifies, manages, communicates and learns from production incidents affecting the Northstar Nexus Platform, Nexus Connect, Nexus Insight and the supporting cloud services in the Mumbai primary region and the Hyderabad secondary (DR) region. It applies to every engineer, engineering manager, Site Reliability Engineering (SRE) member, Customer Success and Support colleague who takes part in an incident, and to anyone who first notices a production problem.

The goals are simple: restore service quickly, keep customers informed in plain language, protect data, and turn every serious incident into durable improvements without blaming individuals. Staffing of the on-call rotation, paging mechanics and personal escalation paths are covered in ENG-005. Alert design and dashboards are covered in ENG-010. Customer-facing handling of tickets, priorities and RCAs is covered in CS-001 and CS-005. Suspected security incidents follow the Security Incident Response process in SEC-010, which runs in parallel with this procedure whenever a breach is suspected.

This procedure covers production only. Problems in development, test or staging environments are handled as normal defects through NorthDesk and do not receive a severity rating.

2. Definitions

- **Incident**: an unplanned event or degradation in production that affects, or is likely to affect, customers or the integrity of customer data.
- **Severity (SEV)**: the internal engineering rating of an incident, from SEV1 (most serious) to SEV4. Severity is about impact and is not the same as the customer-facing priority (P1 to P4) used by Support, although the two are aligned in section 4.
- **Incident Commander (IC)**: the single person who owns the response, makes decisions and delegates work during an incident.

- **Communications Lead**: the person who writes and publishes status updates and coordinates with Customer Success.
- **Operations Lead**: the hands-on technical lead who directs diagnosis and mitigation.
- **Scribe**: the person who keeps a timestamped record of events, decisions and actions.
- **Mitigation**: any action that reduces customer impact, even if the root cause is not yet fixed (for example a rollback, a feature flag switch or a failover).
- **Resolution**: the point at which customer impact has ended and the service is operating within normal parameters.
- **Post-incident review (PIR)**: a blameless written review of an incident that records the timeline, causes, what worked, what did not, and tracked action items.
- **Northstar Status**: the public status page used to inform customers about incidents and maintenance.
- **NorthWatch**: the monitoring, alerting and log search platform that raises most incident alerts.

3. Roles and Responsibilities

Role	Held by	Responsibilities
Incident Commander	On-call engineer or engineering manager trained as IC	Declares severity, assigns roles, sets the communication rhythm, decides on mitigation, declares resolution
Communications Lead	SRE or Customer Success delegate	Publishes updates on Northstar Status, informs Support and account teams, drafts the customer message
Operations Lead	Senior engineer of the affected service	Diagnoses, proposes and executes mitigation, requests extra engineers
Scribe	Any available engineer	Maintains the incident timeline in the NorthDesk incident ticket
Head of Site Reliability Engineering	Procedure owner	Owns this procedure, chairs PIR trend reviews, approves IC training
Engineering manager of the affected service	Service owner	Ensures action items are staffed and completed
Customer Success and Support	CSMs and Support Leads	Handle customer questions, log customer impact, request RCAs

For a SEV1 the Incident Commander must be assigned immediately and must not also act as Operations Lead. For a SEV2 one person may combine IC and Communications Lead for the first hour only if no other trained person is available, and must hand over one of the roles as soon as someone joins. Roles may be handed over during long incidents, but only with an explicit announcement in the incident channel naming the outgoing and incoming person.

4. Severity Definitions and Response Targets

Every production incident must be given a severity within the first minutes of acknowledgement. When in doubt, choose the higher severity and downgrade later; downgrading is free, but a late upgrade costs customers time.

Severity	Definition	Acknowledge	Status updates	Coverage
SEV1	Complete outage, data loss or corruption, or confirmed security breach affecting customers	within 5 minutes	every 30 minutes	24x7, IC assigned immediately
SEV2	Major degradation of a key feature, or multiple customers affected	within 15 minutes	every 60 minutes	24x7 page
SEV3	Minor degradation with a workaround	within 4 business hours	daily	business hours
SEV4	Cosmetic or low impact	next business day	as needed	business hours

4.1 Alignment with customer priorities

SEV1 normally corresponds to customer priority P1 Critical, and SEV2 to P2 High. A single customer reporting a P1 does not automatically create a SEV1; the on-call engineer must first check whether the cause is specific to that customer's configuration. If it is, the ticket remains a customer P1 handled by Support with engineering help, and no SEV is declared unless a second customer is affected.

4.2 Classification guidance

A SEV1 must be declared when any of the following is true: the Nexus Platform API or Pipeline execution is unavailable for all customers in a region; customer production data is lost, corrupted or exposed; authentication fails for most users; or a security breach affecting customers is confirmed. A SEV2 must be declared when a key feature such as Webhooks, Connectors or scheduled Pipelines is badly degraded, when error rates are elevated for several customers, or when a single Enterprise customer loses a core capability and no workaround exists. A SEV3 covers a degraded function with a documented workaround, for example a slow Nexus Insight dashboard that still renders. A SEV4 covers cosmetic defects, typos in the console and non-urgent log noise.

Example: at 14:10 IST NorthWatch alerts that p95 latency for Nexus Connect has tripled and 9 tenants report timeouts on webhook delivery. Three tenants are Enterprise customers. The on-call engineer declares SEV2 at 14:14, within the 15-minute acknowledgement target. At 14:40 the error rate reaches 40% and the Pipeline scheduler stops; the IC upgrades to SEV1, the 5-minute and 30-minute clocks start from the upgrade, and a new IC-plus-Operations-Lead split is applied immediately.

4.3 Changing severity

Only the Incident Commander may change severity. Every change must be recorded in the incident ticket with the reason and the time. When a SEV1 is downgraded to SEV2 the update cadence of the lower severity applies from the next update. An incident must not be downgraded merely because the night is ending or because the team is tired; impact is the only criterion.

5. Detection and Declaration

Incidents are detected through NorthWatch alerts (paging alerts must be customer-impacting symptoms, see ENG-010), Support escalations in NorthAssist, Customer Success reports, internal user reports and synthetic checks. Any employee who suspects a production problem should contact the IT Service Desk on ext. 4000 or raise a NorthDesk ticket of type Production Issue, and

engineers should page the on-call engineer directly through the paging tool described in ENG-005.

5.1 Declaration steps

1 The first responder acknowledges the page within the target for the suspected severity.
2 The first responder checks the NorthWatch golden signals dashboard for the affected service and recent change tickets in NorthDesk.
3 The first responder opens an incident ticket in NorthDesk, sets a provisional severity and creates the incident channel and bridge.
4 For SEV1 and SEV2 the first responder pages the engineering manager on call if no Incident Commander has been assigned.
5 The Incident Commander is confirmed in the channel and states "I am IC" with the time.
6 The Incident Commander assigns the Communications Lead, Operations Lead and Scribe.
7 The Communications Lead publishes the first customer-facing update on Northstar Status within 30 minutes of the SEV1 or SEV2 declaration.

A responder who is unsure about severity must not wait for certainty. Declare, then adjust.

6. Running the Incident

6.1 Priorities

The order of priority is always: protect customer data, restore service, communicate, then diagnose the root cause. Root cause analysis is deliberately delayed until mitigation is complete, unless the cause is needed to choose a mitigation.

6.2 Recent-change check

The first diagnostic question is "what changed?". The Operations Lead must review NorthShip deployments, NorthDesk change tickets, feature flag changes and infrastructure changes in the last 24 hours. If a recent production change plausibly caused the incident, the default decision is to roll back. Per the deployment standard in ENG-003 and ENG-007, an automatic rollback fires when the error rate is above 2% or p95 latency exceeds 1.5x baseline for 5 consecutive minutes, and a rollback must complete within 15 minutes of the decision to roll back. During an incident the IC may order a manual rollback earlier than the automatic trigger.

6.3 Emergency changes during an incident

Emergency fixes follow the Emergency change type: approval by the Engineering Director on call, a NorthDesk change ticket with a rollback plan, and a retrospective within 1 business day. The normal deployment window and Change Advisory Board schedule do not apply to emergency changes, but the rollout stages (canary 10% for 30 minutes, 50% for 30 minutes, then 100%) should still be used unless the IC records a reason to skip them.

6.4 Production access

Engineers needing production access during an incident use the NorthGate bastion, with session recording. Just-in-time privileged access is granted for at most 8 hours; if the incident outlasts that, access must be re-requested. Break-glass accounts may be used only when NorthID is itself unavailable, and every use is reviewed within 1 business day.

6.5 Rhythm and discipline

- The IC speaks in short, explicit decisions and asks for dissent once before deciding.

- Only the Communications Lead speaks to customers or posts to Northstar Status; engineers must not reply to customers in public channels.
- The Scribe records every hypothesis, action, owner and timestamp in the NorthDesk incident ticket, using IST.
- Side investigations are assigned to named people with a stated time-box, typically 15 minutes.
- If the response stalls for more than 30 minutes without progress on a SEV1, the IC must request a second opinion from the engineering manager on call and, if still stalled, escalate per section 8.
- Responders must hand over after about 8 hours of continuous work on a long incident, and the IC must plan relief before fatigue causes errors.

7. Communication

7.1 Customer-facing updates

Customer-facing status updates on Northstar Status must be published within 30 minutes of declaring a SEV1 or SEV2. Customer Success also sends an email to affected Enterprise and Premier contacts within the same 30 minutes, as described in CS-001. Updates must be plain, empathetic and free of blame; they must not contain internal system names, personal data, speculation about the cause or delivery promises that Engineering has not confirmed. Language is English, with Hindi available on request for India customers.

Each update contains four elements: what customers are experiencing, what we know, what we are doing, and when the next update will arrive. The next-update time must be honoured even if there is nothing new to say; a short message such as "We are still working on the mitigation; next update at 16:30 IST" is correct.

7.2 Update cadence

SEV1 updates are published every 30 minutes, and SEV2 updates every 60 minutes. For Enterprise tier customers, Support follows the more frequent customer-communication cadence defined in CS-001 for P1 and P2 tickets. When an incident is resolved, a final "Resolved" update is published, followed by a monitoring period of at least 30 minutes for SEV1 before the incident is closed.

7.3 Internal communication

The incident channel is the single source of truth. The Communications Lead posts a short internal summary every hour for SEV1 to the engineering leadership list, and the COO must be informed of any SEV1 expected to last more than 1 hour. Where a data-loss or breach angle exists, the CISO and the Privacy Officer must be informed at once, and customer notification of a confirmed breach follows SEC-010.

7.4 Example communication

Example: At 02:25 IST, Tier-1 storage latency causes Pipeline runs to fail for all customers in Mumbai. The IC declares SEV1 at 02:28. The Communications Lead publishes "Investigating: some customers are unable to run Pipelines" on Northstar Status at 02:50, within the 30-minute limit. Subsequent updates follow at 03:20, 03:50 and so on, until resolution at 04:10 with a monitoring period until 04:40.

8. Escalation During an Incident

The on-call engineer who does not acknowledge a page within 5 minutes triggers the automatic escalation chain described in ENG-005: secondary on-call, then the engineering manager after a further 10 minutes. During a declared incident the IC may escalate at any time to:

1 The engineering manager of the affected service.
2 The Engineering Director on call.
3 The VP Engineering, for any SEV1 lasting more than 2 hours or any incident with possible data loss.
4 The CTO, for any SEV1 lasting more than 4 hours or any breach with regulatory implications.

Customer Success has its own escalation levels for customer tickets (Level 1 Support Engineer to Level 5 CEO executive sponsor), described in CS-005. These do not replace engineering escalation; the IC should assume that a customer-requested escalation at Level 3 or above will ask for the engineering status and should provide it through the Communications Lead.

9. Closure and Handover

An incident is closed when impact has ended, monitoring shows normal behaviour for the agreed period, a final status update is published, the NorthDesk ticket contains the full timeline, and a PIR owner has been named for SEV1 and SEV2. Incidents that cross shift boundaries require a written handover covering current state, open risks, who is on watch and when the next update is due.

If mitigation is temporary (for example a manual scaling change or a disabled feature), the incident ticket must record the temporary state, the owner, and the date by which it will be reversed or made permanent. Temporary mitigations without an owner and date are not accepted.

10. Post-Incident Review

A blameless post-incident review must be completed within 5 business days of resolution for every SEV1 and SEV2. For SEV3 a short review is recommended when the same service has had repeated incidents.

10.1 PIR contents

1 Summary of customer impact, including start time, end time and affected tenants.
2 Timeline from detection to resolution, taken from the Scribe's record.
3 Contributing factors: technical, process and organisational. The review must not name individuals as causes.
4 What went well and what went badly, including detection delay and communication quality.
5 Action items, each with an owner and a due date, tracked in NorthDesk.
6 Customer-facing RCA status.

10.2 PIR meeting and tracking

The PIR owner (normally the Operations Lead) drafts the review, the IC facilitates a 45-minute meeting, and the Head of Site Reliability Engineering reviews and approves the document. High-priority action items should be completed within 30 days. The Head of Site Reliability Engineering reports open PIR action items monthly to the VP Engineering, and overdue items older than 30 days must be re-justified or re-planned.

10.3 Customer RCA

For a SEV1, the customer-facing RCA is provided to Enterprise and Premier customers within 5 business days of resolution, and to Standard customers on request within 10 business days, as set out in CS-001. The RCA is drafted by Engineering and sent by Customer Success after review. It must be plain, empathetic and blame-free and must not expose internal system names or personal data.

11. Worked Examples

Example 1 (SEV2, no upgrade): A new Connector release causes NX-2004 connector timeouts for 6 tenants at 11:20 IST on a Tuesday. The on-call engineer acknowledges in 6 minutes, the IC declares SEV2, rollback of the canary is decided at 11:35 and completes at 11:48, within 15 minutes of the decision. Status page update is published at 11:45 and the next at 12:45. The PIR is due within 5 business days of resolution.

Example 2 (SEV3): A Nexus Insight dashboard widget is slow for one Premier customer, but data export works. The engineer acknowledges within 4 business hours, posts a daily update in the NorthAssist ticket and no Northstar Status post is needed.

Example 3 (suspected breach): NorthWatch SIEM shows a service account used from an unfamiliar network. The responder raises SEV1 and simultaneously reports to the Security Desk on ext. 5555. The Incident Response Lead from SIRT joins the incident bridge. Containment decisions are shared between the IC and the Incident Response Lead, and forensic evidence is preserved as required by SEC-010.

12. Exceptions and Edge Cases

- Incidents caused by the cloud provider still follow this procedure; customers are informed of impact and workarounds, and the provider's status is referenced without attributing blame.
- A planned maintenance that overruns beyond the announced window becomes an incident and is rated by the actual impact.
- If Northstar Status itself is unavailable, the Communications Lead must use email to Enterprise and Premier contacts and a notice in the Nexus console banner, and open a SEV3 for the status page.
- For incidents spanning a quarter-end freeze or a customer launch freeze, emergency changes remain allowed but need the Engineering Director on call and written notice to the VP Engineering.
- An incident discovered later (for example data corruption found a week after the event) is declared on discovery and the clocks start at discovery time.
- Duplicate incidents for the same cause must be merged under the first ticket, with all customer reports linked.

13. Related Documents

- ENG-005 On-Call and Escalation Handbook: rotations, paging, compensation and escalation timing.
- ENG-010 Observability and Monitoring Standard: alert quality, runbook links and dashboards.
- SEC-010 Security Incident Response: breach handling, forensics and regulatory notifications (regulatory references are illustrative; to be verified by Legal before external use).
- CS-001 Customer Support Handbook: customer priorities, first-response targets and notification rules.

- CS-005 Customer Escalation Procedure: customer-side escalation levels.
- ENG-003 Software Deployment Runbook: rollback and rollout rules.

14. Revision History

Version	Date	Change
2.0	2025-03-10	Introduced Communications Lead role and 30-minute status page rule
2.2	2025-06-18	Clarified severity downgrade rule and customer RCA timings
2.3	2025-09-20	Added edge cases for status page outage and merged duplicate incidents